

Reflective Journal – Module 03

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Reflection on Learning Machine Learning Concepts

During this lab, one of my biggest realizations was how easy it is to misunderstand machine learning concepts when they are only explained theoretically. For example, I initially believed that choosing a more complex model would naturally lead to better results. However, after training both Logistic Regression and a Decision Tree on the same dataset, I saw that the simpler model actually performed better. This challenged my assumption that complexity equals accuracy and helped me understand the importance of matching model choice to the nature of the data.

Another concept that became clearer through practice was overfitting. Before this lab, overfitting felt like an abstract warning rather than a real problem. Seeing how a model can perform well on training data but worse on unseen test data helped me understand that good performance does not automatically mean good learning. The train-test split and evaluation metrics made it clear why validation on unseen data is essential to build trustworthy models.

Understanding Learning Types Through Application

I also gained a deeper understanding of the difference between supervised, unsupervised, and reinforcement learning. While I previously memorized their definitions, applying them to real examples changed how I think about them. The key insight for supervised learning was realizing that it relies on having an “answer key” during training. Without labeled data, supervised learning would not function as intended.

Unsupervised learning became more intuitive once I understood that its goal is not prediction, but discovery. Instead of trying to find the correct answer, the model tries to reveal patterns that may not be obvious to humans. This helped me understand why unsupervised learning is often used in early data exploration or when labels are unavailable.

Reinforcement learning stood out because it does not learn directly from datasets in the same way. Understanding that learning happens through rewards and penalties helped me see why this approach is suitable for environments where decisions influence future outcomes, such as games or robotics.

Model Evaluation and Interpreting Results

One of the most valuable lessons from this lab was learning how to interpret model results beyond accuracy. Initially, I focused mainly on accuracy scores, but the confusion matrix and classification report showed me how misleading accuracy alone can be. These tools helped me understand where the model struggled and how performance varied across different classes, which is critical for real-world applications.

This experience also helped me appreciate why machine learning is an iterative process. Improving a model is not just about rerunning code, but about questioning assumptions, testing alternatives, and understanding the limitations of each approach.

Final Takeaway

Overall, this lab shifted my perspective on machine learning from “running algorithms” to “reasoning about decisions.” I learned that understanding the data, questioning model choices, and carefully evaluating results are just as important as writing correct code. This change in mindset made the machine learning workflow feel less overwhelming and more purposeful, and it gave me confidence to approach more complex problems in future modules.